

Arsitha Sathu

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SUMMARY

Software Engineer with experience in building scalable applications using Java, Spring Boot, and Angular, along with hands-on work in Python-based AI systems. Worked on microservices, REST APIs, and real-time data-driven applications, with experience in PostgreSQL, Kafka, and cloud platforms like AWS and Azure. Focused on writing clean, production-ready code and building reliable backend and full-stack systems.

EDUCATION

MASTER'S DEGREE | Computer Science | George Mason University | GPA – 3.83 Aug 2023 – May 2025
BACHELOR'S DEGREE | Computer Science Engineering | Jawaharlal Technological University | CGPA – 9.21 Aug 2017 – June 2021

TECHNICAL SKILLS

Languages: Java, Python, JavaScript, TypeScript, SQL, C, C++, HTML, CSS, Bash, shell scripting
Frameworks & Libraries: Spring Boot, React.js, Node.js, Angular, FastAPI, PyTorch, OpenAI SDK, LangChain (Agentic Workflows), PyPDF, Spring Security, Hibernate (JPA)
Data & Infrastructure: PostgreSQL, MySQL, MongoDB, REST API, Microservices, WebSockets, AWS (EC2, S3), Azure (AI Foundry, Speech Services, Container Apps, OpenAI (GPT-4o), Azure Key Vault), Docker, Kubernetes, Jenkins, Kafka, CI/CD, RAG, MCP, AudioWorklet
Development Tools: Git, Postman, Swagger, JUnit, JIRA, SonarQube, VS Code, IntelliJ IDEA, ChatGPT, Claude, GitHub Copilot, Cursor, Gemini, Microsoft 365 (Excel, PowerPoint, Teams, Outlook, Word), Mockito
Domain Knowledge: Data Structures & Algorithms, Operating Systems, NLP, LLMs, Data Mining, Networking Fundamentals

PROFESSIONAL EXPERIENCE

BLUE HORSE DIGITAL | Software Engineer | Remote July 2025 – present

- Architected a low-latency, voice agent using Python (aiohttp) and Azure AI Services (Speech/GPT-4o), optimizing client-side audio with a custom JavaScript Audio Worklet and implementing barge-in interruption for sub-second, natural responsiveness.
- Built a custom Agentic Workflow (MCP-based) and Retrieval-Augmented Generation (RAG) engine leveraging Azure OpenAI Embeddings to autonomously execute complex reasoning, query USDA policies, and provide accurate, citation-backed answers.
- Engineered a hands-free PDF Form Filling system that dynamically inspects schemas (using pypdf) to map conversational data into applications in real-time, auto-generating and emailing completed forms without manual input.
- Deployed the scalable, containerized application to Azure Container Apps using Docker multi-stage builds, ensuring secure API key management and robust production infrastructure.

INFOSYS LIMITED | Full-stack Developer | India Oct 2021 – July 2023

- Built backend services using Java 8 and Spring Boot to manage compliance records, audit logs, and regulatory workflows, exposing REST APIs for internal systems.
- Developed responsive UI using Angular and TypeScript, implementing forms and dashboards used by compliance teams to track audit status and policy adherence.
- Designed secure access using Spring Security with role-based authorization, supporting different user roles like auditors and administrators.
- Integrated frontend and backend systems using REST APIs and handled structured data exchange in JSON, ensuring consistency across services.
- Worked with Hibernate and PostgreSQL, writing optimized queries and supporting reporting features for compliance tracking.
- Implemented event-driven updates using Kafka and supported deployments using Docker and Jenkins on AWS, including basic monitoring and production issue handling.

PROJECTS

AI LOG ANALYSIS | Python, React, JavaScript, Spring Boot, GPT-4o-mini, PostgreSQL, Git

- Architected a microservice-based log analysis platform with a Spring Boot backend, a Python FastAPI AI service, and a React frontend for automated error detection and debugging.
- Built a log ingestion pipeline that parses application logs, stores them in PostgreSQL via Spring Data JPA, and triggers AI analysis through WebClient REST integration.
- Developed an interactive React dashboard for one-click log analysis, visualizing GPT-4o-mini generated error clusters, root cause explanations, and actionable fix recommendations.

TASKFLOW | Java 17, Spring Boot, REST APIs, React, Typescript, AWS (SQS, EC2, RDS), PostgreSQL, Docker, Kubernetes, Jenkins, Git

- Built backend services using Java 17/ Spring Boot to handle task creation, updates, and deadline tracking, exposing well-structured APIs for frontend consumption.
- Integrated AWS SQS to process task reminders asynchronously, ensuring notifications are triggered without blocking user requests.
- Containerized services using Docker and deployed them on Kubernetes, configuring deployments and services for scalable execution.
- Set up a Jenkins CI/CD pipeline to automate build, test, Docker image creation, and deployment to a Kubernetes cluster hosted on AWS EC2, with data stored in AWS RDS (PostgreSQL).

ABSTRACTIVE TEXT SUMMARIZATION | Python, mT5 Transformer, Git, PyTorch, Transformers, Wandb, Pandas, NumPy

- Fine-tuned an mT5 Transformer on a custom dataset of 200+ Telugu narratives, implementing a specialized NLP preprocessing pipeline and achieving optimized performance tracked via ROUGE metrics and Wandb.

DEPARTMENTAL CONNECT | ReactJS, HTML, CSS, Bootstrap, JavaScript, Node.js, SQL, Git

- Developed a dynamic full-stack application using React, Node.js, and MySQL, resulting in a 70% improvement in load times and a 45% increase in user engagement.

ACHIEVEMENTS AND ACTIVITIES

- Received Outstanding Academic Achievement Award during my master's in computer science at George Mason University.
- Won 1st place in the CSE Department Webathon technical competition during my bachelor's.
- Held responsibilities as a Department Magazine Co-Ordinator and class representative during my undergraduate studies for 4 years.

CERTIFICATIONS

- AWS Certified Cloud Practitioner